

Multiple Choice

1. **Answer:** B
Options: A) commutative semi group; B) abelian group; C) non-abelian group; D) None of these
Note: Correct option: B - abelian group
2. **Answer:** B
Options: A) not anti-symmetric; B) transitive; C) reflexive; D) symmetric
Note: Correct option: B - transitive
3. **Answer:** D
Options: A) 0; B) 1; C) 0,1; D) 0,4
Note: Correct option: D - 0,4
4. **Answer:** D
Options: A) finite; B) cyclic; C) of order two; D) abelian
Note: Correct option: D - abelian
5. **Answer:** B
Options: A) 6; B) 12; C) 30; D) 105
Note: Correct option: B - 12

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: ' $x^2 + y^2 = 9$ '.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The set of all functions from $\mathbb{R}$ to $\{0, 1\}$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** II only. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.